Nuclei: Composition, Size, Binding Energy & Nuclear Reactions

1. Learning Objectives and Syllabus Orientation

In the National Eligibility cum Entrance Test (NEET UG), the chapter **Nuclei** contributes predictable, high-scoring questions. These questions test both conceptual clarity (such as the properties of nuclear forces, saturation, and the binding energy curve) and quick numerical calculations (such as nuclear radii, mass defect, energy released in fission and fusion, and Q-value).

Core NEET Target Competencies:

- Understand the composition of the nucleus, discovery of the neutron, and definitions of isotopes, isobars, and isotones.
- Master the empirical nuclear radius formula $R = R_0 A^{1/3}$ and understand why nuclear matter density is independent of mass number A.
- Define mass defect ΔM, atomic mass unit (u), and calculate binding energy E_b and binding energy per nucleon E_{bn}.
- Interpret the binding energy per nucleon versus mass number curve, its peaks, saturation behavior, and physical rationale for fission and fusion.
- Analyze the fundamental properties of the strong nuclear force and the nucleon-nucleon potential energy diagram.
- Calculate the Q-value of nuclear reactions and understand stellar thermonuclear fusion cycles (proton-proton cycle).

2. Composition of the Nucleus and Basic Constituents

Rutherford's α-particle scattering experiments established that an atom consists of a tiny, extremely dense, positively charged core called the **nucleus**, surrounded by atomic electrons. The linear size of a nucleus is roughly $10^{-14}\ \text{m}$, which is approximately 10^4 times smaller than the radius of an atom ($\sim 10^{-10}\ \text{m}$). Consequently, the volume of the nucleus is only about 10^{-12} of the atomic volume, yet it accommodates more than 99.9% of the total atomic mass.

2.1 The Atomic Mass Unit (u)

Because atomic and nuclear masses are minuscule in kilograms (for instance, one ^{12}C atom has a mass of $1.992647 \times 10^{-26}\ \text{kg}$), physicists use the unified **atomic mass unit (u)**. It is defined as exactly $\frac{1}{12}$th of the mass of an unexcited neutral carbon-12 atom:

DEFINITION OF ATOMIC MASS UNIT

$$1\ \text{u} = \frac{\text{Mass of one } ^{12}\text{C atom}}{12} = \frac{1.992647 \times 10^{-26}\ \text{kg}}{12} \approx 1.660539 \times 10^{-27}\ \text{kg}$$

$$1\ \text{u} \times c^2 \approx 931.5\ \text{MeV}$$

2.2 Constituents of the Nucleus: Protons and Neutrons

The nucleus contains two types of fundamental particles, collectively designated as **nucleons**:

- **Proton (p):** Discovered as the nucleus of the lightest hydrogen atom (^1_1H). It carries a fundamental positive charge of $+e = +1.602 \times 10^{-19}\ \text{C}$ and has a rest mass:

$$m_p = 1.00727\ \text{u} = 1.67262 \times 10^{-27}\ \text{kg}$$

- **Neutron (n):** Discovered by **James Chadwick in 1932** by bombarding beryllium ($^{9}\mathrm{Be}$) with α-particles:
$$^{4}_{2}\alpha + ^{9}_{4}\mathrm{Be} \rightarrow ^{12}_{6}\mathrm{C} + ^{1}_{0}n$$
The emitted neutral radiation knocked out high-speed protons from light nuclei such as paraffin wax, helium, and nitrogen. Chadwick proved via simultaneous conservation of linear momentum and energy that this penetrating radiation consisted of neutral particles with a mass almost equal to that of the proton, rather than ultra-energetic γ-photons.
$$m_n = 1.00866\ \mathrm{u} = 1.67493 \times 10^{-27}\ \mathrm{kg}$$

> **Key NEET Point (Neutron Stability):** An isolated, free neutron is unstable and undergoes β^--decay into a proton, an electron, and an antineutrino:
> $$n \rightarrow p + e^- + \bar{\nu}$$
> with a mean life of about $1000\ \mathrm{s}$ (half-life $\approx 14\ \mathrm{minutes}$). Inside a stable nucleus, however, the neutron is completely stable due to binding interactions.

2.3 Nuclear Terminology and Classification of Nuclides

A specific nuclear species or **nuclide** is symbolized as $^{A}_{Z}\mathrm{X}$:

- **Atomic Number (Z):** Number of protons in the nucleus. This defines the chemical element and its position in the periodic table. Nuclear charge $= +Ze$.
- **Neutron Number (N):** Number of neutrons in the nucleus.
- **Mass Number (A):** Total number of nucleons: $A = Z + N$.

Classification	Defining Condition	Key Characteristics	Illustrative Examples
Isotopes	Same Z, different A (different N)	Identical chemical properties; identical position in periodic table; different masses.	$^{1}_{1}\mathrm{H}$ (Protium), $^{2}_{1}\mathrm{H}$ (Deuterium), $^{3}_{1}\mathrm{H}$ (Tritium); $^{35}_{17}\mathrm{Cl}$ and $^{37}_{17}\mathrm{Cl}$
Isobars	Same A, different Z and N	Different chemical elements; different physical and chemical behaviors; identical nucleon total.	$^{3}_{1}\mathrm{H}$ and $^{3}_{2}\mathrm{He}$; $^{14}_{6}\mathrm{C}$ and $^{14}_{7}\mathrm{N}$
Isotones	Same $N = (A - Z)$, different Z and A	Different elements; equal number of neutrons.	$^{197}_{79}\mathrm{Au}$ ($N = 118$) and $^{198}_{80}\mathrm{Hg}$ ($N = 118$); $^{13}_{6}\mathrm{C}$ ($N = 7$) and $^{14}_{7}\mathrm{N}$ ($N = 7$)

Worked Example: Fractional Atomic Mass from Isotopic Abundance

Natural chlorine is composed of two stable isotopes: $^{35}\mathrm{Cl}$ with mass $34.98\ \mathrm{u}$ and relative abundance 75.4%, and $^{37}\mathrm{Cl}$ with mass $36.98\ \mathrm{u}$ and relative abundance 24.6%. Calculate the average atomic mass of chlorine.

Step-by-Step Solution:

The average atomic mass is the weighted average of its constituent isotopic masses:
$$\bar{m} = \frac{(75.4 \times 34.98) + (24.6 \times 36.98)}{100} = \frac{2637.492 + 909.708}{100} = \frac{3547.20}{100} \approx 35.47\ \mathrm{u}$$
This weighted average accounts for why the atomic mass of chlorine in the periodic table is non-integral ($\approx 35.5\ \mathrm{u}$).

BOSONANCE

3. Nuclear Size and Invariance of Nuclear Density

Rutherford's α-scattering measurements showed that the distance of closest approach for a $5.5\ \text{MeV}$ α-particle head-on collision with a gold nucleus ($Z = 79$) is approximately $4.0 \times 10^{-14}\ \text{m}$, setting an upper bound for the nuclear radius. Subsequent high-energy electron scattering experiments provided an accurate measure of nuclear charge distribution and nuclear radii.

NUCLEAR RADIUS FORMULA

$$R = R_0 A^{1/3}$$

where $R_0 \approx 1.2\ \text{fm} = 1.2 \times 10^{-15}\ \text{m}$, and A is the mass number of the nuclide.

3.1 Nuclear Volume and Nuclear Density

Assuming the nucleus to be spherical of radius R, its volume is:

$$V = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi\left(R_0 A^{1/3}\right)^3 = \frac{4}{3}\pi R_0^3 A$$

Thus, the nuclear volume is directly proportional to the mass number A ($V \propto A$).

The mass of a nucleus containing A nucleons is approximately $M \approx A \cdot m_N$, where $m_N \approx 1.66 \times 10^{-27}\ \text{kg}$ is the average mass of a nucleon. The density of nuclear matter ρ_{nucl} is:

$$\rho_{\text{nucl}} = \frac{\text{Mass}}{\text{Volume}} = \frac{A \cdot m_N}{\frac{4}{3}\pi R_0^3 A} = \frac{m_N}{\frac{4}{3}\pi R_0^3}$$

Crucial NEET Takeaway: Nuclear Density is Constant!

- The mass number A cancels out identically in the numerator and denominator.
- Nuclear density is **independent of the mass number** A and atomic number Z.
- All nuclei, from hydrogen to uranium, have approximately the same density:

$$\rho_{\text{nucl}} \approx \frac{1.66 \times 10^{-27}}{\frac{4}{3}\pi(1.2 \times 10^{-15})^3} \approx 2.3 \times 10^{17}\ \text{kg/m}^3$$

- This immense density ($2.3 \times 10^{14}\ \text{g/cm}^3$) is comparable only to the density of **neutron stars**, indicating that nuclear matter is tightly packed like an incompressible liquid drop.

Worked Example: Ratio of Nuclear Radii

Find the approximate ratio of the nuclear radius of gold $^{197}_{79}\text{Au}$ to that of silver $^{107}_{47}\text{Ag}$.

Step-by-Step Solution:

Using the relation $R \propto A^{1/3}$:

$$\frac{R(\text{Au})}{R(\text{Ag})} = \left(\frac{A_{\text{Au}}}{A_{\text{Ag}}}\right)^{1/3} = \left(\frac{197}{107}\right)^{1/3} \approx (1.841)^{1/3} \approx 1.225 \approx 1.23$$

The radius of the gold nucleus is only about 23% larger than that of silver, despite having nearly double the mass, because radius scales with the cube root of mass number.

4. Mass-Energy Equivalence, Mass Defect & Binding Energy

4.1 Einstein's Mass-Energy Equivalence

Einstein's special theory of relativity states that mass and energy are mutually convertible:

$$E = \Delta m \cdot c^2$$

For a mass of $1\ \text{g} = 10^{-3}\ \text{kg}$, the energy equivalent is:

$$E = (10^{-3}\ \text{kg}) \times (3 \times 10^{8}\ \text{m/s})^2 = 9 \times 10^{13}\ \text{J}$$

For $1\ \text{u}$:

$$E = (1.660539 \times 10^{-27}\ \text{kg}) \times (2.997925 \times 10^{8}\ \text{m/s})^2 \approx 1.4924 \times 10^{-10}\ \text{J} \approx 931.5\ \text{MeV}$$

4.2 Mass Defect (ΔM)

High-precision mass spectrometry shows that the rest mass of any stable bound nucleus M is always strictly **less** than the sum of the masses of its constituent free protons and neutrons.

MASS DEFECT EQUATION

$$\Delta M = [Z \cdot m_p + (A - Z) \cdot m_n] - M_{\text{nucleus}}$$

When expressed in terms of neutral atomic masses $m(A, Z)$ and the hydrogen atomic mass $m_{\text{H}} = m_p + m_e$:

$$\Delta M = [Z \cdot m_{\text{H}} + (A - Z) \cdot m_n] - m({}^{A}_{Z}\text{X})$$

where the electron masses cancel out precisely to an excellent approximation.

4.3 Nuclear Binding Energy (E_b) and Binding Energy per Nucleon (E_{bn})

The mass defect does not disappear; it represents the energy liberated when individual, free nucleons combine to form the bound nucleus. Conversely, it is the energy that must be supplied from outside to break a nucleus completely into its isolated, non-interacting constituent nucleons.

$$E_b = \Delta M \cdot c^2 = \Delta M(\text{in u}) \times 931.5\ \text{MeV}$$

A more direct measure of nuclear stability is the **binding energy per nucleon (E_{bn})**:

$$E_{bn} = \frac{E_b}{A}$$

The larger the value of E_{bn}, the more tightly bound and energetically stable the nucleus is.

Worked Example: Binding Energy Calculation for Oxygen-16

Calculate the mass defect, total binding energy, and binding energy per nucleon of ${}^{16}_{8}\text{O}$.
Given data: Mass of ${}^{16}_{8}\text{O}$ atom $= 15.99493\ \text{u}$, mass of proton $m_p = 1.00727\ \text{u}$, mass of neutron $m_n = 1.00866\ \text{u}$, and mass of electron $m_e = 0.00055\ \text{u}$.

Step-by-Step Solution:

1. Mass of 8 protons: $8 \times 1.00727\ \text{u} = 8.05816\ \text{u}$
2. Mass of 8 neutrons: $8 \times 1.00866\ \text{u} = 8.06928\ \text{u}$

Total mass of free constituents $= 8.05816 + 8.06928 = 16.12744\ \text{u}$

3. Nuclear mass of ${}^{16}_{8}\text{O} = 15.99493 - (8 \times 0.00055) = 15.99493 - 0.00440 = 15.99053\ \text{u}$
4. Mass defect ΔM:

$$\Delta M = 16.12744\ \text{u} - 15.99053\ \text{u} = 0.13691\ \text{u}$$

5. Total binding energy E_b:

$$E_b = 0.13691 \times 931.5\ \text{MeV} \approx 127.53\ \text{MeV}$$

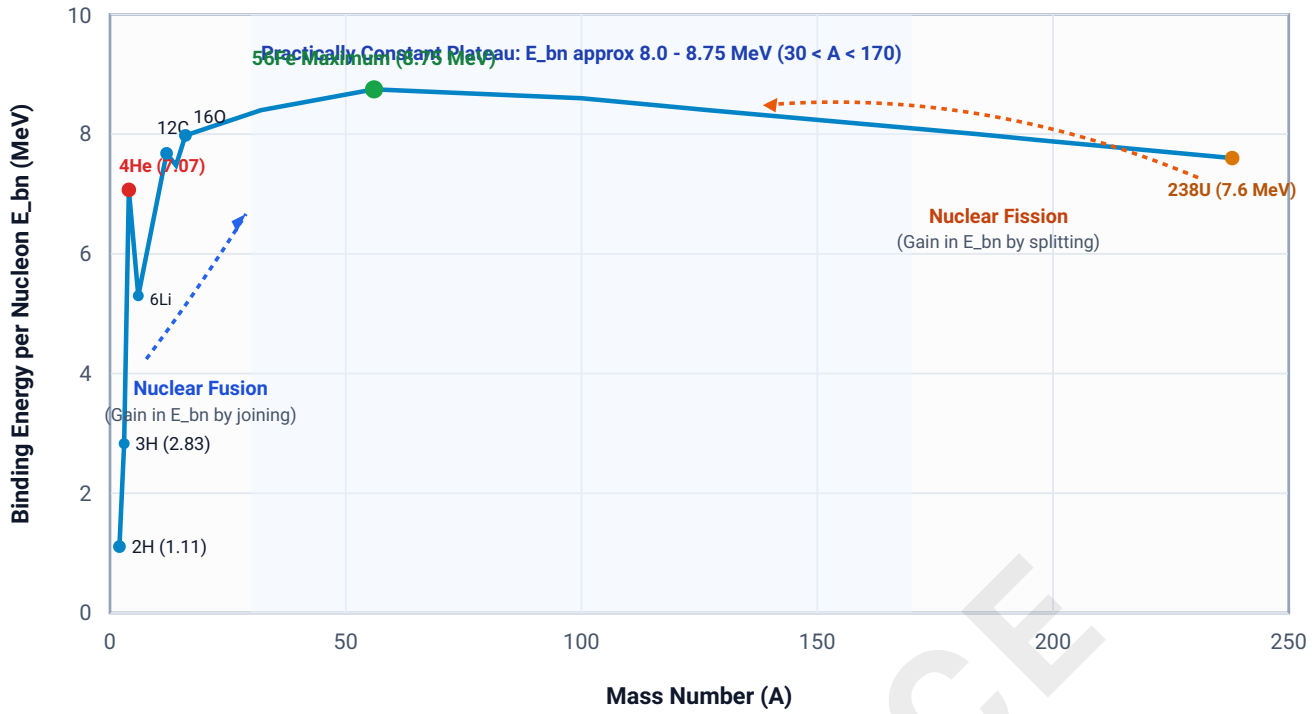
6. Binding energy per nucleon E_{bn}:

$$E_{bn} = \frac{127.53\ \text{MeV}}{16} \approx 7.97\ \text{MeV/nucleon}$$

5. The Binding Energy per Nucleon Curve and Its Physical Significance

The plot of binding energy per nucleon E_{bn} versus mass number A summarizes the relative stability of all known nuclides and explains the physical origin of nuclear energy release.

BOSONANCE



Tutorial Diagram 1: Binding energy per nucleon (E_{bn}) as a function of mass number A, showing the prominent plateau, the maximum at $^{56}\mathrm{Fe}$, and the energetic pathways for fusion and fission.

5.1 Cardinal Features of the E_{bn} vs A Curve

1. **Constancy in the Intermediate Mass Region ($30 < A < 170$):** The curve is virtually flat, exhibiting values between $8.0\ \mathrm{MeV}$ and $8.75\ \mathrm{MeV}$ per nucleon. The absolute maximum occurs at $^{56}_{26}\mathrm{Fe}$ ($\approx 8.75\ \mathrm{MeV/nucleon}$), making iron-56 one of the most stable nuclei in nature.
2. **Drop at Heavy Nuclei ($A > 170$):** For heavy nuclei, E_{bn} gradually decreases due to increasing Coulomb repulsion among the large number of protons. For $^{238}_{92}\mathrm{U}$, E_{bn} drops to approximately $7.6\ \mathrm{MeV/nucleon}$.
3. **Drop and Fine Structure at Light Nuclei ($A < 30$):** E_{bn} is substantially lower for light nuclei (e.g., $1.11\ \mathrm{MeV}$ for $^{2}_{1}\mathrm{H}$). However, distinct local peaks appear at $^{4}_{2}\mathrm{He}$ ($7.07\ \mathrm{MeV}$), $^{12}_{6}\mathrm{C}$ ($7.68\ \mathrm{MeV}$), and $^{16}_{8}\mathrm{O}$ ($7.98\ \mathrm{MeV}$). These peaks reflect the enhanced stability of nuclei composed of complete α-clusters or closed nuclear shells.

5.2 Two Fundamental Conclusions: Why Fission and Fusion Release Energy

- **Nuclear Fission:** When a heavy, less tightly bound nucleus (such as $A \approx 240$ with $E_{bn} \approx 7.6\ \mathrm{MeV}$) fragments into two intermediate nuclei (such as $A \approx 120$ with $E_{bn} \approx 8.5\ \mathrm{MeV}$), the binding energy per nucleon increases by:

  $$\Delta E_{bn} \approx 8.5 - 7.6 = 0.9\ \mathrm{MeV/nucleon}$$

  The total energy liberated in one such fission event is:

  $$Q = A \times \Delta E_{bn} \approx 240 \times 0.9\ \mathrm{MeV} \approx 216\ \mathrm{MeV} \approx 200\ \mathrm{MeV}$$
- **Nuclear Fusion:** When very light nuclei ($A \leq 10$) with small E_{bn} combine to form a heavier nucleus (such as $^{4}_{2}\mathrm{He}$ with $E_{bn} \approx 7.07\ \mathrm{MeV}$), the binding energy per nucleon jumps dramatically. Because the daughter nucleus is significantly more tightly bound than the reactants, massive energy is released per unit mass.

5.3 The Saturation Property of Nuclear Forces

Why does E_{bn} remain constant over such a broad range of A instead of increasing proportionally to A?

If the nuclear force were long-ranged (like Coulomb or gravitational forces), each nucleon would interact with all other $(A-1)$ nucleons. The total binding energy would then be proportional to $A(A-1) \approx A^2$, and $E_{bn} = E_b/A$ would increase linearly with A.

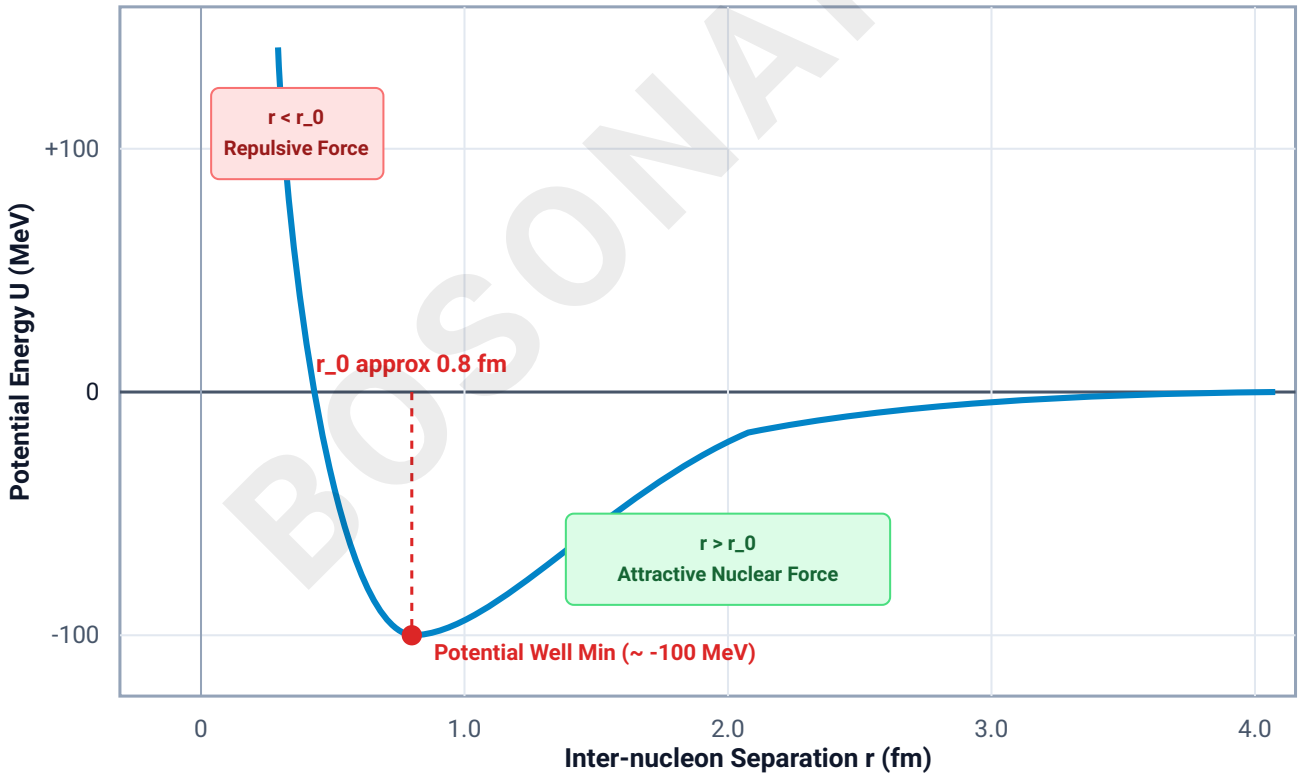
BOSONANCE

In reality, the strong nuclear force has a **short range** (≈ 1 to 2 fm). A given interior nucleon interacts only with its immediate neighbors within this short range (say, a maximum of p neighbors). Its binding contribution is thus fixed at $p \cdot k$ (where k is an interaction constant). Adding more nucleons increases the nuclear volume, but does not alter the local environment of interior nucleons. This localized interaction is termed the **saturation property of the nuclear force**.

6. Nature and Properties of the Nuclear Force

The force binding nucleons within the tiny dimensions of the nucleus is the **strong nuclear force**. Its defining properties are:

1. **Strongest Fundamental Force:** Within its range, it is roughly 100 times stronger than the electrostatic Coulomb repulsion between protons, and 10^{38} times stronger than gravitational attraction.
2. **Short Range:** It operates effectively only over distances of a few femtometres (≤ 2--3 fm) and vanishes rapidly beyond that.
3. **Charge Independence:** The nuclear force between two protons, two neutrons, or a proton and a neutron is virtually identical, provided Coulomb repulsion is excluded:
$$F_{nn} \approx F_{np} \approx F_{pp} \text{ (nuclear part)}$$
4. **Non-Central and Spin-Dependent:** Unlike Coulomb's law or Newton's law of gravitation, the nuclear force is not purely central and depends on the relative spin orientations of the interacting nucleons.
5. **Repulsive Core at Extremely Short Distances:** Prevents the nucleus from collapsing in on itself.



Tutorial Diagram 2: Potential energy of a nucleon pair as a function of separation r. The curve reaches a pronounced minimum at $r_0 \approx 0.8$ fm, corresponding to stable equilibrium.

6.1 Interpreting the Potential Energy Curve

The force $F(r)$ between two nucleons is given by the negative gradient of potential energy:

$$F(r) = -\frac{\mathrm{d}U}{\mathrm{d}r}$$

- **Equilibrium Separation ($r = r_0 \approx 0.8$ fm):** At this point, $\frac{\mathrm{d}U}{\mathrm{d}r} = 0$, meaning the net force is zero. The potential energy reaches its minimum depth of approximately -100 MeV.

- **Repulsive Zone ($r < r_0$):** The slope $\frac{\mathrm{d}U}{\mathrm{d}r} < 0$, so $F(r) > 0$ (repulsive force). This extremely hard repulsive core prevents nucleons from coalescing together and maintains finite nuclear density.
- **Attractive Zone ($r > r_0$):** The slope $\frac{\mathrm{d}U}{\mathrm{d}r} > 0$, so $F(r) < 0$ (attractive force). As r increases beyond $\approx 3\text{--}4\ \mathrm{fm}$, $U(r) \to 0$ asymptotically, confirming the short-range nature.

7. Nuclear Stability and Radioactivity Overview

Natural radioactivity was discovered accidentally by **A. H. Becquerel in 1896** while investigating uranium compounds wrapped in light-proof black paper. Subsequent investigations proved that radioactivity is a spontaneous nuclear phenomenon whereby an unstable parent nucleus transforms into a more stable daughter nucleus by emitting particles or electromagnetic radiation.

7.1 The Neutron-to-Proton Ratio (N/Z) for Stability

- For light nuclei ($Z \leq 20$), stability requires $N/Z \approx 1$. Examples: ${}_{2}^{4}\mathrm{He}$, ${}_{6}^{12}\mathrm{C}$, ${}_{8}^{16}\mathrm{O}$.
- For heavy nuclei, electrostatic repulsion among numerous protons increases rapidly (scaling as Z^2). To maintain cohesion, the nucleus requires excess neutrons to provide additional attractive strong force without contributing electric charge. The stability ratio gradually climbs to $N/Z \approx 1.5$ (or $3:2$) near bismuth and lead.
- Nuclei with N/Z significantly higher or lower than this stability band undergo radioactive decay. In fact, only roughly 10% of all known nuclides are stable.

7.2 The Three Modes of Radioactive Decay

Decay Mode	Emitted Entity	Transmutation Equation	Change in Z and A
α-decay	Helium nucleus ${}_{2}^{4}\mathrm{He}$	${}_{Z}^{A}\mathrm{X} \to {}_{Z-2}^{A-4}\mathrm{Y} + {}_{2}^{4}\mathrm{He}$	Z decreases by 2, A decreases by 4
β^--decay	Electron (e^-) & antineutrino ($\bar{\nu}$)	${}_{Z}^{A}\mathrm{X} \to {}_{Z+1}^{A}\mathrm{Y} + e^- + \bar{\nu}$	Z increases by 1, A remains unchanged
β^+-decay	Positron (e^+) & neutrino (ν)	${}_{Z}^{A}\mathrm{X} \to {}_{Z-1}^{A}\mathrm{Y} + e^+ + \nu$	Z decreases by 1, A remains unchanged
γ-decay	High-energy photon (γ)	${}_{Z}^{A}\mathrm{X}^* \to {}_{Z}^{A}\mathrm{X} + \gamma$	Z and A remain unchanged

8. Nuclear Reactions, Q-Value, Fission, and Fusion

8.1 Nuclear Reactions and the Q-Value

In any nuclear reaction represented by $A + b \to C + d + Q$, the following quantities are strictly conserved:

- Total electric charge ($\sum Z$)
- Total number of nucleons or baryon number ($\sum A$)
- Total relativistic mass-energy
- Linear momentum and angular momentum

REACTION ENERGY (Q-VALUE)

$$Q = (\text{Initial rest mass} - \text{Final rest mass}) \times c^2 = [(m_A + m_b) - (m_C + m_d)]c^2$$
$$Q = (\text{Final kinetic energy}) - (\text{Initial kinetic energy})$$

- **Exothermic Reaction ($Q > 0$):** Rest mass is converted into kinetic energy of products and radiation. The reaction can occur spontaneously for stationary targets if energetic and tunneling criteria are met.
- **Endothermic Reaction ($Q < 0$):** Kinetic energy is converted into rest mass. The reaction cannot occur unless the incident projectile supplies threshold kinetic energy.

BOSONANCE

8.2 Nuclear Fission

Fission involves the splitting of a heavy nucleus into two medium-mass fragments following the absorption of a thermal (slow) neutron:

$$ {}^{1}_{0}n + {}^{235}_{92}\mathrm{U} \rightarrow {}^{236}_{92}\mathrm{U}^{*} \rightarrow {}^{144}_{56}\mathrm{Ba} + {}^{89}_{36}\mathrm{Kr} + 3\,{}^{1}_{0}n + Q $$

Other typical fission channels include:

$$ {}^{1}_{0}n + {}^{235}_{92}\mathrm{U} \rightarrow {}^{133}_{51}\mathrm{Sb} + {}^{99}_{41}\mathrm{Nb} + 4\,{}^{1}_{0}n $$
$$ {}^{1}_{0}n + {}^{235}_{92}\mathrm{U} \rightarrow {}^{140}_{54}\mathrm{Xe} + {}^{94}_{38}\mathrm{Sr} + 2\,{}^{1}_{0}n $$

Energetics of Fission:

- Average energy released per fission event is approximately $200\ \mathrm{MeV}$.
- The bulk of this energy (≈ 80--85%) appears as kinetic energy of the heavy fission fragments. The remainder appears as kinetic energy of prompt neutrons, γ-rays, and subsequent β-decay daughter activity.
- Fission of $1\ \mathrm{kg}$ of ${}^{235}\mathrm{U}$ releases $\approx 10^{14}\ \mathrm{J}$ of energy, compared to only $\approx 10^{7}\ \mathrm{J}$ from burning $1\ \mathrm{kg}$ of coal (a factor of 10^{7} difference).

8.3 Nuclear Fusion and Stellar Energy (The Proton-Proton Cycle)

When light nuclei fuse into a heavier nucleus, substantial energy is released because the product has a higher binding energy per nucleon.

The Coulomb Barrier: Because both reacting nuclei are positively charged, they experience strong electrostatic repulsion. To come within the range of the nuclear attractive force (≈ 1--$2\ \mathrm{fm}$), they must overcome a potential barrier. For two protons, the barrier height is:

$$ U_C = \frac{1}{4\pi\varepsilon_0}\frac{e^2}{r} \approx 400\ \mathrm{keV} $$

The thermal kinetic energy required to overcome this barrier classically is $\frac{3}{2}kT \approx 400\ \mathrm{keV}$, requiring a temperature of:

$$ T \approx \frac{2 \times (400 \times 10^{3} \times 1.6 \times 10^{-19})}{3 \times (1.38 \times 10^{-23})} \approx 3 \times 10^{9}\ \mathrm{K} $$

Inside the solar core, the temperature is $1.5 \times 10^{7}\ \mathrm{K}$. Thermonuclear fusion takes place here because protons in the high-energy Maxwellian tail possess kinetic energies well above the average, and quantum mechanical tunneling allows them to penetrate the Coulomb barrier.

THE SOLAR PROTON-PROTON (P-P) CYCLE

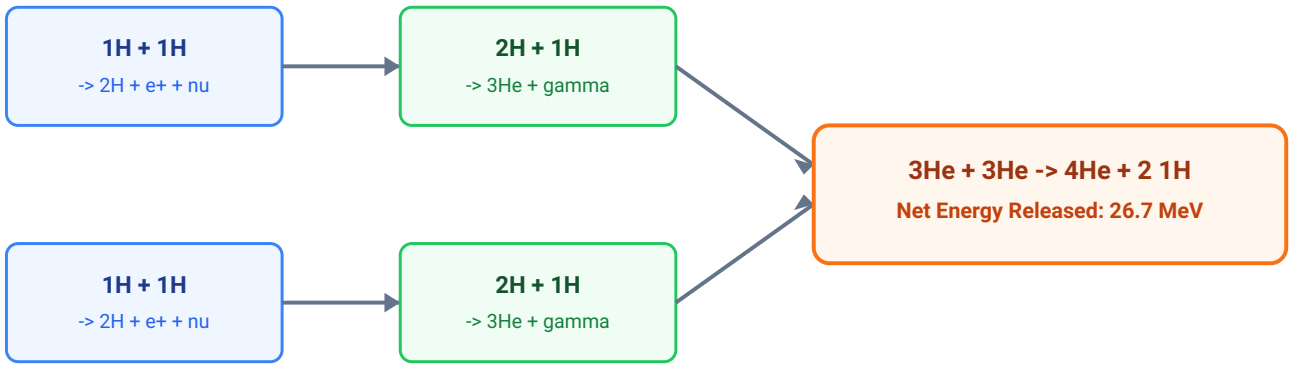
The multi-step reaction cycle powering our Sun is:

$$ \text{(i)}\quad {}^{1}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{2}_{1}\mathrm{H} + e^{+} + \nu + 0.42\ \mathrm{MeV} $$
$$ \text{(ii)}\quad e^{+} + e^{-} \rightarrow \gamma + \gamma + 1.02\ \mathrm{MeV} \quad (\text{annihilation}) $$
$$ \text{(iii)}\quad {}^{2}_{1}\mathrm{H} + {}^{1}_{1}\mathrm{H} \rightarrow {}^{3}_{2}\mathrm{He} + \gamma + 5.49\ \mathrm{MeV} $$
$$ \text{(iv)}\quad {}^{3}_{2}\mathrm{He} + {}^{3}_{2}\mathrm{He} \rightarrow {}^{4}_{2}\mathrm{He} + 2\,{}^{1}_{1}\mathrm{H} + 12.86\ \mathrm{MeV} $$

Combining $2 \times (\mathrm{i}) + 2 \times (\mathrm{ii}) + 2 \times (\mathrm{iii}) + (\mathrm{iv})$, the net reaction is:

$$ 4\,{}^{1}_{1}\mathrm{H} + 2\,e^{-} \rightarrow {}^{4}_{2}\mathrm{He} + 2\nu + 6\gamma + 26.7\ \mathrm{MeV} $$

BOSONANCE



Tutorial Diagram 3: Schematic representation of the solar proton-proton chain yielding a $^4_2\mathrm{He}$ nucleus and $26.7\ \mathrm{MeV}$ of total energy.

8.4 Chemical vs. Nuclear Reactions: Conceptual Distinction

Property	Chemical Reaction (e.g., $2\mathrm{H}_2 + \mathrm{O}_2 \rightarrow 2\mathrm{H}_2\mathrm{O}$)	Nuclear Reaction (e.g., $^1_0n + {}^{235}_{92}\mathrm{U} \rightarrow \ldots$)
Conservation of Elements	Strictly conserved; atoms are only rearranged into new molecular combinations.	Not conserved; elements are transmuted into entirely different chemical species.
Conservation of Protons & Neutrons	Separately conserved as whole atomic entities.	Total charge number Z and total mass number A are conserved.
Typical Energy Scale	A few electron-volts (eV) per reaction event.	Mega electron-volts (MeV) per event (a factor of 10^6 larger).
Mass Defect Conversion	Occurs in principle ($\Delta m = \Delta E/c^2$), but $\Delta m \sim 10^{-35}\ \mathrm{kg}$ is unmeasurably small.	Readily measurable mass defect directly convertible to kinetic energy via $E = \Delta m \cdot c^2$.

9. NEET UG Practice Questions (With Complete Worked Solutions)

The following questions reflect the pattern, cognitive depth, and calculation style of the NEET UG examination. Each question includes four options, the correct answer, and an exhaustive step-by-step explanation.

BOSONANCE

Practice Question 1 [Nuclear Radius Scaling]

If the nuclear radius of $^{27}_{13}\mathrm{Al}$ is $3.6\ \mathrm{fm}$, then the approximate nuclear radius of $^{125}_{52}\mathrm{Te}$ is:

A. $4.8\ \mathrm{fm}$
B. $6.0\ \mathrm{fm}$
C. $7.2\ \mathrm{fm}$
D. $9.6\ \mathrm{fm}$

Correct Answer: B ($6.0\ \mathrm{fm}$)

Worked Solution:

The radius of a nucleus is related to its mass number A by $R = R_0 A^{1/3}$.

$$\frac{R(\mathrm{Te})}{R(\mathrm{Al})} = \left(\frac{A_{\mathrm{Te}}}{A_{\mathrm{Al}}}\right)^{1/3} = \left(\frac{125}{27}\right)^{1/3} = \frac{5}{3}$$

$$R(\mathrm{Te}) = \frac{5}{3} \times R(\mathrm{Al}) = \frac{5}{3} \times 3.6\ \mathrm{fm} = 5 \times 1.2\ \mathrm{fm} = 6.0\ \mathrm{fm}$$

Practice Question 2 [Nuclear Density Ratio]

The mass numbers of two nuclei are in the ratio $1:8$. The ratio of their respective nuclear matter densities is:

A. $1:8$
B. $1:2$
C. $1:1$
D. $8:1$

Correct Answer: C ($1:1$)

Worked Solution:

Nuclear mass is $M \approx A \cdot m_N$ and nuclear volume is $V = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi R_0^3 A$. Therefore, the nuclear density is:

$$\rho = \frac{M}{V} = \frac{A \cdot m_N}{\frac{4}{3}\pi R_0^3 A} = \frac{m_N}{\frac{4}{3}\pi R_0^3}$$

Because ρ is independent of the mass number A, the ratio of nuclear densities for any two nuclei is identically $1:1$.

BOSONANCE

Practice Question 3 [Potential Energy and Nuclear Force Direction]

The potential energy of a pair of nucleons is minimum at a separation $r_0 \approx 0.8\ \text{fm}$. For an inter-nucleon separation of $r = 0.5\ \text{fm}$, the nuclear force between them is:

A. Strongly attractive
B. Strongly repulsive
C. Zero
D. Purely electrostatic and repulsive

Correct Answer: B (Strongly repulsive)

Worked Solution:

The force is given by $F = -\frac{\mathrm{d}U}{\mathrm{d}r}$. At $r_0 \approx 0.8\ \text{fm}$, the potential energy $U(r)$ has an absolute minimum ($\frac{\mathrm{d}U}{\mathrm{d}r} = 0$).

For separations $r < r_0$ (here $r = 0.5\ \text{fm}$), the potential energy curve rises extremely steeply, giving a negative slope ($\frac{\mathrm{d}U}{\mathrm{d}r} < 0$). Hence, $F = -\left(-\left|\frac{\mathrm{d}U}{\mathrm{d}r}\right|\right) > 0$, corresponding to a powerful repulsive force that prevents nuclear collapse.

Practice Question 4 [Fission Energy Calculation]

A heavy nucleus of mass number $A = 240$ has a binding energy per nucleon of $7.6\ \text{MeV}$. It undergoes complete fission into two equal fragments, each having a mass number of $A = 120$ and a binding energy per nucleon of $8.5\ \text{MeV}$. The total energy released during the fission of this single nucleus is:

A. $108\ \text{MeV}$
B. $200\ \text{MeV}$
C. $216\ \text{MeV}$
D. $180\ \text{MeV}$

Correct Answer: C ($216\ \text{MeV}$)

Worked Solution:

1. Total initial binding energy of the parent nucleus ($A = 240$):
$$E_{b,\text{initial}} = 240 \times 7.6\ \text{MeV} = 1824\ \text{MeV}$$
2. Total final binding energy of the two fragments ($2 \times 120 = 240$ nucleons):
$$E_{b,\text{final}} = 2 \times (120 \times 8.5\ \text{MeV}) = 240 \times 8.5\ \text{MeV} = 2040\ \text{MeV}$$
3. The energy released is the gain in binding energy:
$$Q = E_{b,\text{final}} - E_{b,\text{initial}} = 2040\ \text{MeV} - 1824\ \text{MeV} = 216\ \text{MeV}$$
Alternatively: $Q = A \times \Delta E_{bn} = 240 \times (8.5 - 7.6)\ \text{MeV} = 240 \times 0.9 = 216\ \text{MeV}$.

BOSONANCE

Practice Question 5 [Assertion-Reason: Saturation of Nuclear Force]

Assertion (A): The binding energy per nucleon is practically constant in the mass number range $30 < A < 170$.
Reason (R): The nuclear force is short-ranged and exhibits the saturation property, meaning a nucleon interacts only with its immediate neighbors.

A. Both (A) and (R) are true and (R) is the correct explanation of (A).
B. Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
C. (A) is true but (R) is false.
D. (A) is false but (R) is true.

Correct Answer: A (Both are true and R explains A)

Worked Solution:

In a medium-sized or large nucleus, each nucleon is surrounded by a limited number of nearest neighbors determined by the short range of the nuclear force (≈ 1--$2\ \text{fm}$). Any nucleon situated farther away exerts negligible nuclear force. Consequently, adding more nucleons to the nucleus increases the volume proportionally but leaves the binding energy contributed per nucleon inside constant ($\approx pk$). Hence, the saturation property directly accounts for the broad constancy of E_{bn}.

BOSONANCE

Practice Question 6 [Energy Released from Fuel Mass in Fission]

Assuming that an average of $180\ \text{MeV}$ of energy is released per fission of $^{239}_{94}\text{Pu}$, how much total energy (in MeV) is released if $1.0\ \text{kg}$ of pure ^{239}Pu undergoes complete fission? (Take Avogadro's number $N_A = 6.02 \times 10^{23}\ \text{mol}^{-1}$)

A. $4.53 \times 10^{23}\ \text{MeV}$
B. $4.53 \times 10^{26}\ \text{MeV}$
C. $7.55 \times 10^{25}\ \text{MeV}$
D. $2.48 \times 10^{24}\ \text{MeV}$

Correct Answer: B ($4.53 \times 10^{26}\ \text{MeV}$)

Worked Solution:

1. Number of moles in $1000\ \text{g}$ of ^{239}Pu:

$$n = \frac{1000\ \text{g}}{239\ \text{g/mol}} \approx 4.184\ \text{mol}$$

2. Number of nuclei N:

$$N = n \times N_A = 4.184 \times 6.02 \times 10^{23} \approx 2.519 \times 10^{24}\ \text{nuclei}$$

3. Total energy released:

$$E_{\text{total}} = N \times 180\ \text{MeV} = 2.519 \times 10^{24} \times 180\ \text{MeV} \approx 4.53 \times 10^{26}\ \text{MeV}$$

Practice Question 7 [Coulomb Barrier of Deuterons]

Two deuterons ($^2_1\mathrm{H}$) undergo a head-on collision. Assuming each deuteron is a hard sphere of radius $R = 2.0\ \mathrm{fm}$, what is the minimum kinetic energy required to overcome their mutual electrostatic barrier when they just touch each other? ($\frac{1}{4\pi\varepsilon_0} = 9 \times 10^9\ \mathrm{N \cdot m^2/C^2}$, $e = 1.6 \times 10^{-19}\ \mathrm{C}$)

A. $180\ \mathrm{keV}$
B. $360\ \mathrm{keV}$
C. $720\ \mathrm{keV}$
D. $90\ \mathrm{keV}$

Correct Answer: B ($360\ \mathrm{keV}$)

Worked Solution:

When the two deuterons just touch, the separation between their centers is:

$$r = R_1 + R_2 = 2.0\ \mathrm{fm} + 2.0\ \mathrm{fm} = 4.0\ \mathrm{fm} = 4.0 \times 10^{-15}\ \mathrm{m}$$

The electrostatic potential energy at this contact distance is:

$$U = \frac{1}{4\pi\varepsilon_0}\frac{q_1 q_2}{r} = \frac{(9 \times 10^9) \times (1.6 \times 10^{-19})^2}{4.0 \times 10^{-15}}$$

$$U = \frac{9 \times 10^9 \times 2.56 \times 10^{-38}}{4.0 \times 10^{-15}} = \frac{23.04 \times 10^{-29}}{4.0 \times 10^{-15}} = 5.76 \times 10^{-14}\ \mathrm{J}$$

Converting Joules to keV:

$$U = \frac{5.76 \times 10^{-14}\ \mathrm{J}}{1.6 \times 10^{-19}\ \mathrm{J/eV}} = 3.6 \times 10^5\ \mathrm{eV} = 360\ \mathrm{keV}$$

BOSONANCE

Practice Question 8 [Energy Released in Deuterium Fusion and Lamp Lifetime]

Consider the fusion reaction: ${}_1^2\mathrm{H} + {}_1^2\mathrm{H} \rightarrow {}_2^3\mathrm{He} + {}_0^1n + 3.27\ \mathrm{MeV}$. How long can a $100\ \mathrm{W}$ electric lamp be kept glowing by the complete fusion of $2.0\ \mathrm{kg}$ of deuterium? (Take $1\ \mathrm{year} \approx 3.15 \times 10^7\ \mathrm{s}$)

A. $5.0 \times 10^4\ \mathrm{years}$

B. $4.9 \times 10^4\ \mathrm{years}$

C. $1.57 \times 10^5\ \mathrm{years}$

D. $3.14 \times 10^3\ \mathrm{years}$

Correct Answer: B ($4.9 \times 10^4\ \mathrm{years}$)

Worked Solution:

1. Mass of deuterium $m = 2000\ \mathrm{g}$. Molar mass $= 2\ \mathrm{g/mol}$.

Number of deuterium atoms: $N = \frac{2000}{2} \times 6.023 \times 10^{23} = 6.023 \times 10^{26}\ \mathrm{nuclei}$.

2. Since two deuterium nuclei participate in each fusion reaction, the number of fusion events is:

$$N_{\mathrm{reactions}} = \frac{N}{2} = 3.0115 \times 10^{26}$$

3. Total energy released:

$$E_{\mathrm{total}} = 3.0115 \times 10^{26} \times 3.27\ \mathrm{MeV} = 9.848 \times 10^{26}\ \mathrm{MeV}$$
$$E_{\mathrm{total}} = 9.848 \times 10^{26} \times (1.6 \times 10^{-13}\ \mathrm{J}) \approx 1.576 \times 10^{14}\ \mathrm{J}$$

4. Power of the lamp $P = 100\ \mathrm{W} = 100\ \mathrm{J/s}$. The glowing time t is:

$$t = \frac{E_{\mathrm{total}}}{P} = \frac{1.576 \times 10^{14}}{100} = 1.576 \times 10^{12}\ \mathrm{s}$$

5. In years:

$$t = \frac{1.576 \times 10^{12}\ \mathrm{s}}{3.154 \times 10^7\ \mathrm{s/year}} \approx 4.99 \times 10^4 \approx 5.0 \times 10^4\ \mathrm{years}$$

BOSONANCE

Practice Question 9 [Spontaneous Fission Feasibility of Iron-56]

Consider the hypothetical symmetric fission of a ${}^{56}_{26}\mathrm{Fe}$ nucleus into two identical ${}^{28}_{13}\mathrm{Al}$ fragments:

$$ {}^{56}_{26}\mathrm{Fe} \rightarrow 2\ {}^{28}_{13}\mathrm{Al} $$

Given: $m({}^{56}_{26}\mathrm{Fe}) = 55.93494\ \mathrm{u}$ and $m({}^{28}_{13}\mathrm{Al}) = 27.98191\ \mathrm{u}$. Which statement is correct?

A. $Q = +26.90\ \mathrm{MeV}$; the fission is energetically highly favorable and spontaneous.
B. $Q = -26.90\ \mathrm{MeV}$; the fission is endothermic and cannot occur spontaneously.
C. $Q = +13.45\ \mathrm{MeV}$; the fission proceeds readily with slow neutrons.
D. $Q = -4.32\ \mathrm{MeV}$; the reaction is exothermic but Coulomb suppressed.

Correct Answer: B ($Q = -26.90\ \mathrm{MeV}$; endothermic and impossible spontaneously)

Worked Solution:

1. Mass of reactants: $m_{\text{initial}} = 55.93494\ \mathrm{u}$
2. Mass of products: $m_{\text{final}} = 2 \times 27.98191\ \mathrm{u} = 55.96382\ \mathrm{u}$
3. Calculate the Q-value:

$$ Q = [m_{\text{initial}} - m_{\text{final}}] \times 931.5\ \mathrm{MeV} = [55.93494 - 55.96382] \times 931.5\ \mathrm{MeV} $$

$$ Q = -0.02888\ \mathrm{u} \times 931.5\ \mathrm{MeV/u} \approx -26.90\ \mathrm{MeV} $$

Because $Q < 0$, the process requires an input of $26.90\ \mathrm{MeV}$ and cannot occur spontaneously. This aligns with ${}^{56}\mathrm{Fe}$ sitting at the apex of the binding energy curve.

Practice Question 10 [Matching Classification of Nuclides]

Match List-I with List-II and select the correct code:

List-I (Nuclide Pair)	List-II (Classification)
(A) ${}^{1}_{1}\mathrm{H}$ and ${}^{2}_{1}\mathrm{H}$	1. Isobars
(B) ${}^{3}_{1}\mathrm{H}$ and ${}^{3}_{2}\mathrm{He}$	2. Isotones
(C) ${}^{197}_{79}\mathrm{Au}$ and ${}^{198}_{80}\mathrm{Hg}$	3. Isotopes
(D) ${}^{238}_{92}\mathrm{U}$ and ${}^{235}_{92}\mathrm{U}$	4. Nuclear Isomers

A. A-3, B-1, C-2, D-3
B. A-1, B-3, C-2, D-4
C. A-3, B-2, C-1, D-3
D. A-2, B-1, C-3, D-4

Correct Answer: A (A-3, B-1, C-2, D-3)

Worked Solution:

- (A) ${}^{1}_{1}\mathrm{H}$ and ${}^{2}_{1}\mathrm{H}$: Same atomic number $Z = 1$, different mass number $A \rightarrow$ **Isotopes** (3).
- (B) ${}^{3}_{1}\mathrm{H}$ and ${}^{3}_{2}\mathrm{He}$: Same mass number $A = 3$, different $Z \rightarrow$ **Isobars** (1).
- (C) ${}^{197}_{79}\mathrm{Au}$ ($N = 197 - 79 = 118$) and ${}^{198}_{80}\mathrm{Hg}$ ($N = 198 - 80 = 118$): Same neutron number $N = 118 \rightarrow$ **Isotones** (2).
- (D) ${}^{238}_{92}\mathrm{U}$ and ${}^{235}_{92}\mathrm{U}$: Same atomic number $Z = 92 \rightarrow$ **Isotopes** (3).

BOSONANCE

10. Common Student Pitfalls & Exam Checklist

High-Frequency NEET Mistakes:

- **Confusing R with ρ:** Nuclear radius scales as $A^{1/3}$, whereas nuclear density is *independent* of A. If a problem asks for the ratio of densities of $^{12}\mathrm{C}$ and $^{238}\mathrm{U}$, the answer is $1:1$, not $(12/238)$ or $(12/238)^{1/3}$.
- **Forgetting Electron Mass Cancellation:** In mass defect calculations involving $\Delta M = [Zm_{\mathrm{H}} + (A-Z)m_n] - m_{\mathrm{atom}}$, using atomic masses automatically cancels the Z atomic electron masses.
- **Equating Equilibrium Point with Force Zero:** In the potential energy curve $U(r)$, at $r_0 \approx 0.8\ \mathrm{fm}$, the potential energy is at its *minimum* ($-100\ \mathrm{MeV}$), which means the *force* is zero ($F = -\mathrm{d}U/\mathrm{d}r = 0$). At $r < 0.8\ \mathrm{fm}$, the force is repulsive; at $r > 0.8\ \mathrm{fm}$, it is attractive.
- **Free Neutron vs. Bound Neutron:** Remember that a free neutron is unstable (β^- decays with mean life $\sim 1000\ \mathrm{s}$), while inside a stable nucleus it is completely stable.
- **Number of Fusion Reactions vs. Number of Deuterons:** In deuterium fusion ($^2\mathrm{H} + {}^2\mathrm{H} \rightarrow \ldots$), each reaction consumes **two** deuterons. The total number of reactions is $N/2$.

Rapid NEET Revision Checklist

- ✓ $1\ \mathrm{u} = 1.660539 \times 10^{-27}\ \mathrm{kg} \iff 931.5\ \mathrm{MeV}/c^2$.
- ✓ $R = R_0 A^{1/3}$ with $R_0 \approx 1.2\ \mathrm{fm}$.
- ✓ $\rho_{\mathrm{nucl}} \approx 2.3 \times 10^{17}\ \mathrm{kg/m^3}$ (constant for all nuclei).
- ✓ Mass defect: $\Delta M = [Zm_p + (A-Z)m_n] - M_{\mathrm{nucleus}}$.
- ✓ Binding energy: $E_b = \Delta M \cdot c^2$; Binding energy per nucleon: $E_{bn} = E_b/A$.
- ✓ Maximum E_{bn} is at $^{56}\mathrm{Fe}$ ($8.75\ \mathrm{MeV/nucleon}$); $^{238}\mathrm{U}$ is $\approx 7.6\ \mathrm{MeV/nucleon}$.
- ✓ Nuclear forces: short-ranged, charge-independent, strongest in nature, exhibits saturation.
- ✓ Equilibrium separation: $r_0 \approx 0.8\ \mathrm{fm}$ (attractive for $r > r_0$, repulsive for $r < r_0$).
- ✓ Fission of $A = 240 \rightarrow 2 \times 120$ releases ≈ 200--$216\ \mathrm{MeV}$ ($\approx 0.9\ \mathrm{MeV/nucleon}$).
- ✓ Solar proton-proton cycle net reaction: $4\ {}^1_1\mathrm{H} + 2e^- \rightarrow {}^4_2\mathrm{He} + 2\nu + 6\gamma + 26.7\ \mathrm{MeV}$.

11. Final Summary Recap

The nucleus is a dense central structure containing Z protons and N neutrons held together by the strong nuclear force. The nuclear radius increases with mass number according to $R \propto A^{1/3}$, which ensures that all nuclei have an identical, constant density of approximately $2.3 \times 10^{17}\ \mathrm{kg/m^3}$.

The stability of a nucleus is governed by its binding energy per nucleon E_{bn}. The E_{bn} versus A curve features a broad plateau around $8.5\ \mathrm{MeV}$ between $A = 30$ and $A = 170$ with a peak at $^{56}\mathrm{Fe}$ ($8.75\ \mathrm{MeV}$), dropping off for both light and heavy nuclei. This shape explains why **nuclear fission** of heavy elements and **nuclear fusion** of light elements are both exothermic processes. Nuclear forces operate over short ranges (≈ 1--$2\ \mathrm{fm}$), display saturation, and become repulsive below $0.8\ \mathrm{fm}$, maintaining stability across the periodic table.

BOSONANCE